

Mohammad Amirifard

AI/ML Engineer

Contact

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Professional Summary

A Machine Learning Engineer with **3+ years** of experience in **Generative AI**, **Large Language Models**, and **Computer Vision**, pursuing an MSc at Politecnico di Milano. Delivered **15+ AI projects** across academia and industry, including internships at **STMicroelectronics** and workshops at **Nokia**. Skilled in ML/DL, LLMs, and scalable model deployment, with proven results such as improving model accuracy by 40% and accelerating AI benchmarking by 10x.

Skills

Programming: Python, MATLAB

ML/DL Frameworks: TensorFlow, Pytorch, Keras, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib

Cloud and MLOps: AWS, Azure ML Studio, Linux, Git, GitHub, Docker, Kubernetes, SQL, FastAPI

Others: English(C1), Italian(A2), Teamwork and Organization, Creativity

Work Experience

Machine Learning Engineer

STMicroelectronics Company, Internship

05/2025 - Present

Milan, Italy

- Designed and deployed automated **neural architecture search (NAS)** workflows for **LLM agent bots**, improving model accuracy and computational efficiency by 40%.
- Implemented **zero-cost proxy** optimization methods, accelerating **generative AI bench-marketing** by 10X, while lowering resource consumption.
- Constructed and integrated an **AutoNN LLM agent**, that automatically designs neural network structures, reducing model architecture generation time 70%.

Machine Learning Engineer

Polimi Data Scientists Association, Part-time

02/2023 - 10/2024

Milan, Italy

- Developed, trained, and optimized 10+** machine learning models in Computer vision, anomaly detection and time series.
- Organised data **pipelines** for efficient **ETL** processes and source relevant datasets, adopted in 5+ projects like SIIT and Breast Cancer detection.
- Showcased project progress to team members and international students biweekly, successfully leading to the completion of **4 significant projects**, using **GenAI**.

Data Scientist

Doran Smart-City, Full-time

09/2021 - 09/2022

Tehran, Iran

- Executed machine learning algorithms like **Random Forest** and **XGBoost** to analyze and model urban traffic data, resulting in a **15% improvement in traffic flow**.
- Redesigned predictive models using gradient boosting and random forests for smart energy management, leading to a **10% reduction in energy consumption** within city infrastructure.

Education

Polytechnic University of Milan

M.Sc. Telecommunication Engineering – Computer Science and communications

09/2022 - Present

Milan, Italy

Amirkabir University of Technology

B.Sc. Engineering

09/2016 - 09/2021

Tehran, Iran

Projects

- Long Form Question Answering ChatBot** | Retrieval-augmented generation (RAG), LLAMA, LoRA, BART, NLP, Hugging Face
- Created an NLP pipeline for Long Form Question Answering using transformers (BART, LLAMA) and RAG in a team of 4, securing a ROUGE score of 22.
- Zero-Cost Proxies in AutoNN LLM Agent** | Google Cloud, Azure ML Studio, AutoML, GenAI, Fine Tuning, Gemini, OpenAI
- Integrated zero-cost proxies into AutoNN LLM agent, achieving 0.78 Kendall's Tau on CIFAR-x datasets for reliable training-free architecture performance estimation.
- SIIT ChatBot** | GPT, BERT, LLAMA, Hugging Face, Pytorch, FAISS
- Spearheaded a technical team of 10 in the SIIT project with JEMP startup for SIIT Company, revolutionizing pharmaceutical synthesis and slashing production time by 25% by developing a chatbot.
- Cooperative Perception In Automated Cars** | 3D Object detection, GenAI, Transformers
- Enhanced 3D object detection, like , in cooperative scenarios by optimizing PointPillar-based models on the OPV2V dataset (232k annotated objects), leveraging generative AI techniques.
- Traffic Forecasting** | Time Series, Pattern Recognition, Decision trees, RNN, LSTM, Conv Nets
- Directed a team of 3 to achieve a 9% MAPE in estimating 50 Mbits/s channels using GEANT Dataset with RNN and CONV-LSTM networks.
- Other Projects** | R-CNN, Yolo, Mask learning, Generative AI, Semantic Segmentation, data science
- 1) ClimateFish Clustering, 2) Mac-Derandomization, 3)Application Identifier, 4)Breast Cancer Detection,
 - 5) ML Intrusion Detection, 6) Univariate Time Series Forecasting, 7) Plant Classification (Healthy vs Unhealthy)

Workshop Experience

- Impact of non-Gaussian noise on LDPC performance Project** 06/2024
Nokia Company Vimercate, Italy
- Presented to over 100 attendees at **Nokia**, showcasing a simulation of non-Gaussian noise effects on LDPC performance using QAM 4, 16, and 64.
- MultiNet-Route-Optimizer Project** 03/2024
Nokia Company Vimercate, Italy
- Presented a **65% improvement in routing efficiency for a Nokia-defined problem**, achieved by a 4-person team using Top-Down and Bottom-Up optimization strategies.

Certifications

- Neural Networks and Deep Learning** | DeepLearning.AI
- Supervised Learning** | From DeepLearning.AI
- Linux LPIC 1** | Tehran Institute of Technology
- Python** | Tehran Institute of Technology